Indian Institute of Information Technology Dharwad

Large Language Models

Mid Semester Examination

Timing: 9.30 AM - 10.30 PM | 7th Semester | Full Marks: 20

No marks will be awarded if all parts of a question are not answered together or if question/sub-question numbers are wrong/missing

1. Answer the following questions in brief: 4X2=8
   a. What is the key limitation of traditional n-gram language models?
   The key limitation is its fixed, limited context window, which prevents it from capturing long-range dependencies in text.
   b. Why is positional encoding required in transformers?
   To provide word-order/positional information, since self-attention alone does not inherently encode the sequence order of tokens.
   c. What is the difference between encoder-only and decoder-only architectures?
   Encoder-only models use bidirectional attention to understand input context, mainly for tasks like classification and representation learning; decoder-only models use causal (left-to-right) attention to predict the next token, mainly for text generation.
   d. What is masked attention and why is it used in GPT-like models?
   Masked attention prevents a token from attending to future tokens, ensuring causal, left-to-right prediction. It is used in GPT-like models to prevent information leakage during next-token prediction.

2. Given the following:
   i. Query (Q) = [1, 0]
   ii. Key (K1) = [1, 1], Key (K2) = [0, 1]
   iii. Value (V1) = [2, 0], Value (V2) = [0, 3]

   a) Compute attention scores using dot product (Q·K)
   $Q \cdot K1 = (1\times1) + (0\times1) = 1$
   $Q \cdot K2 = (1\times0) + (0\times1) = 0$
   Attention scores = [1, 0]
   b) Apply softmax to obtain attention weights (approximate)
   Softmax([1, 0]) = [0.731, 0.269]
   c) Compute final output vector 2+2+2
   Final output = $(0.731 \times [2, 0]) + (0.269 \times [0, 3]) = [1.462, 0.807]$

3. Consider a language model predicting next-word probabilities:

i. Word A: 0.4
ii. Word B: 0.3
iii. Word C: 0.2
iv. Word D: 0.1

a) What word is selected using greedy decoding?

Word A, because it has the highest probability (0.4).

b) If top-k (k=2) is used, which words are considered?

Word A (0.4) and Word B (0.3).

c) Briefly explain how top-p differs from top-k. 3

Top-k selects a fixed number (k) of highest-probability tokens, whereas top-p (nucleus sampling) selects the smallest set of tokens whose cumulative probability reaches p.

4. For a training corpus: "I like NLP," "I like AI," "I love NLP," use a bigram language model and Add-1 (Laplace) smoothing to compute probability for: "I like ML." Add <S> and </S> tokens. 3

With <S> and </S>:

Training corpus:

- <S> I like NLP </S>
- <S> I like AI </S>
- <S> I love NLP </S>

Vocabulary size $V = 8$: {<S>, </S>, I, like, love, NLP, AI, ML}

Using Add-1 smoothing:

$P(I| < S >) = \frac{3+1}{3+8} = \frac{4}{11}$

$P(like|I) = \frac{2+1}{3+8} = \frac{3}{11}$

$P(ML|like) = \frac{0+1}{2+8} = \frac{1}{10}$

$P(< /S > |ML) = \frac{0+1}{0+8} = \frac{1}{8}$

Therefore,

$P(IlikeML) = \frac{4}{11} \times \frac{3}{11} \times \frac{1}{10} \times \frac{1}{8}$

P("I like ML")=3/2420 = 0.00124